School-Based Autism Identification:

Prevalence, Racial Disparities, and Systemic Correlates

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Abstract

This study examined prevalence and relative risk of school-based autism identification by race, and the relations of each to state characteristics. The sample was drawn from general and special education enrollment data for students ages 6 to 21 in 46 states during the 2008-2009 academic year. The results show that 1 in 228 students nationwide was identified with autism for special education eligibility, but there was substantial variability across states, with some states nine times more likely to identify students with autism than others. There were significant differences in prevalence between racial groups at the state level. Hispanic and American Indian/Alaskan Native students were often less likely to be identified with autism relative to White students whereas Asian/Pacific Islander students were commonly more likely to be identified with autism than White students. Regression analysis indicated systemic predictors of prevalence and minority relative risk varied.

Background

- The CDC currently estimates 1 in 88 children has an autism spectrum disorder (ASD). These disorders are characterized by deficits in:
  - communication skills
  - social skills
  - repetitive or restricted behaviors or interests
- Students with ASD may be entitled to special education. Rapid increases in ASD prevalence present challenges for school systems lacking the resources and infrastructure to provide appropriate services.
- Racial disparities are common in other areas of special education, but disparities in school-based autism treatment are largely unexplored.
- There is some evidence of racial disparities in clinical diagnoses of autism.
  - CDC's surveillance study showed that minority children were less likely to have documented ASD diagnoses than White children.
  - Clinicians take longer to identify autism in Black children than white children.

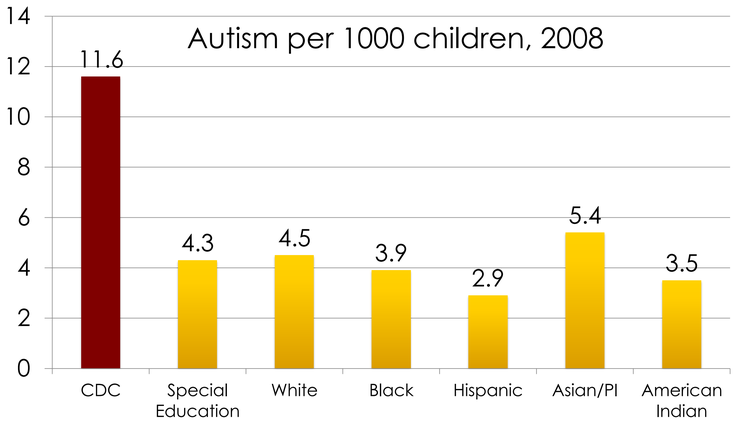
Research Questions

1. **To what extent does prevalence of school-based autism vary across states?**
2. **To what extent does relative risk of autism diagnosis vary among the four race/ethnicity minority groups, compared to White students, at the national and state levels?**
3. **How are prevalence and relative risk predicted by state population demographics and special education policy?**

Autism defined under special education law: "a developmental disability significantly affecting verbal and nonverbal communication and social interaction, generally evident before age three, that adversely affects a child's educational performance. Other characteristics often associated with autism are engagement in repetitive activities and stereotyped movements, resistance to environmental change or change in daily routines, and unusual responses to sensory experiences." (§ 300.8(c)(1))

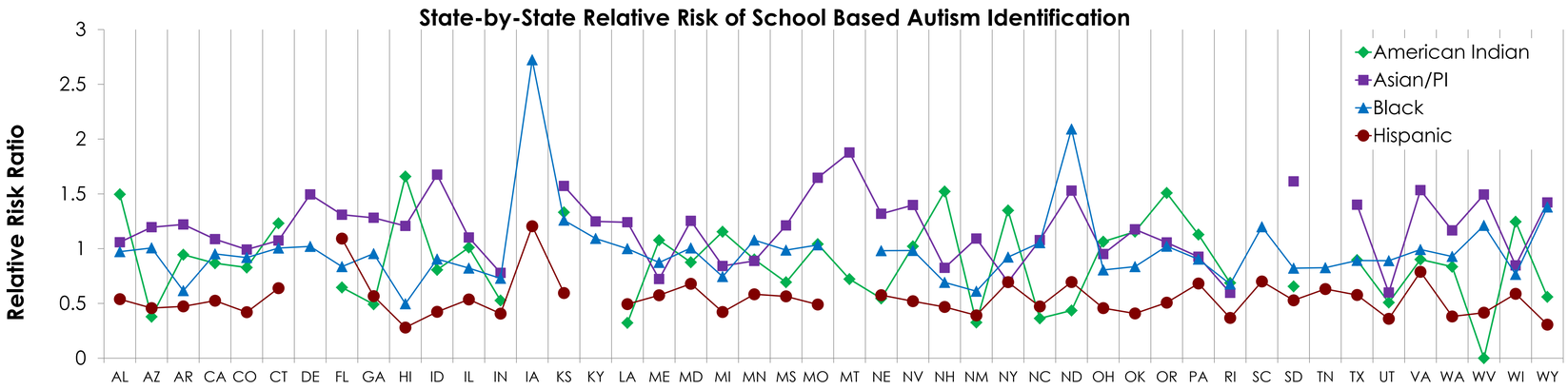
Method

- Data Source: 2008 state enrollment data submitted by departments of education; 2008 Census estimates
- Sample: 46 states reporting enrollment data for 65.9 million students.
- Analyses
  - Prevalence in school-age population
  - Prevalence by race
  - Predicted prevalence and racial disparities by state characteristics: state enrollment; minority enrollment; size of special education program; prevalence of related disorders; change in diagnoses; state autism policy



Results

- Nationally, 1 in 228 students were identified with autism in schools.
- Across states, prevalence ranged from 1 in 1000 to 1 in 101.
- Racial disparities were common.
  - Black students were less likely than Whites to receive services for autism in 9 states.
  - Hispanic students were less likely than Whites to receive services for autism in 44 states.
  - American Indian students were less likely than Whites to receive services for autism in 21 states.
  - Asian students were more likely than Whites to receive services for autism in 21 states.
- Overall prevalence was predicted by size of state special education programming, prevalence of related disorders, and state autism policy.



Racial minority groups' relative risk of school-based autism identification varied substantially by state in 2008-2009. The relative risk ratio is a ratio of the prevalence for the minority group and White students. In states where a groups' marker is absent (e.g., American Indian and Asian/Pacific Islander students in South Carolina), this indicates there was insufficient subgroup population to calculate relative risk. In most states (*n*=33), Asian students were more likely to be identified with autism than Whites as indicated by relative risk ratios above 1 (indicated by the dotted horizontal line). Hispanics students tended to be less likely to be identified as states' relative risk ratios were commonly less than 1 (*n*=41). Although the highest relative risk was found among Black students in Iowa and North Dakota, Black students were less likely to be identified than White students in 30 states.

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